Moe Rahnema

Contact: mrahnema@neurowit.net

mrahnema@hotmail.com

linkedin.com/in/moe-rahnema-5713195

Currently: Residing in SE Asia

Master of Science and Engineers degrees from MIT and Northeastern university in engineering sciences (Aerospace, electronics) and electrical engineering with Ph.D equivalent coursework

About:

Technical Consultant and Trainer in mobile telecommunications, data analytics, and AI technologies and applications

Current Interests:

My current interests include training in AI algorithms, and applications, system analysis, AI startups and Investment, and product design

I am currently involved in independent research and honorary training in AI, traveling, and exploring related investment opportunities

AI Experience:

Researched the application of unsupervised machine learning techniques for mobile network performance analysis, trend forecasting, and optimization, for contributing to improved network planning methodologies. Published research outcomes in the book *Network Planning and Optimization of GSM and UMTS Networks* (Wiley, 2008).

Conducted extensive research on deep learning algorithms, focusing on model architecture evaluation, performance optimization, and practical deployment strategies. Designed and delivered specialized AI and deep learning training programs in the form of PowerPoint presentations, listed in the following, for various related workshops and seminars in West Asia

- A Comprehensive Guide to Variational AutoEncoders: “Fundamentals, Mathematical Framework, and Implementation Strategies", Nov 2024 for workshop presentation

- · The Evolution of AI with Diffusion Models: Theoretical Concepts, Mechanisms, and Applications, Oct 2024, for workshop presentation
- · Boltzaman Machine Learning Analysis and Case Studies, Oct 2024, prepared for seminar presentation
- · RNN Gradient Problems and the LSTM Solution and Analysis, Oct 2024,prepared for seminar presentation
- · A Review of Artificial Intelligence Concepts, Technologies and Applications, January 2024, prepared for seminar presentation
- · Practical Aspects in the Design and Training of Neural Networks, Sept 2024, prepared for training and seminar presentation
- · AI-Driven Data Analysis: Unsupervised Learning Techniques and Case Studies, July 2024, prepared for training and seminar presentation

- · Modeling Chaotic Systems: Python Simulations of Double Pendulum, Lorentz Convection Model, and Population Growth Dynamics, May 2020, prepared for training and seminar presentation
- · Magnetic Resonance Imaging: Principles, Acquisition, and Interpretation, February 2024, prepared for training and seminar presentation

Previous Work History/Experience

Served as a Technical Consultant across the United States, Russia, Mexico, China, Malaysia, Singapore, and Indonesia, collaborating with leading telecommunications organizations including Nokia, Ericsson, Maxis, Huawei, LCC, Consistel, Exel, and Axis Telecommunications. Developed and implemented methodologies for the planning, analysis, and optimization of cellular networks, while serving as Project Manager and Lead Technical Consultant responsible for recruiting, building, mentoring, and leading high-performance engineering teams. Successfully directed RF optimization initiatives that significantly improved network coverage and KPI performance in Jakarta, Malaysia, Mexico City, and Russia, consistently achieving high levels of customer satisfaction and effectively communicating technical challenges and strategic solutions to executive management. These practical experiences were later incorporated into my book, Network Planning and Optimization of GSM and UMTS Networks, published by Wiley in 2008. Prior to 2000, served as Principal Engineer at Motorola and Hughes Network Systems, where I developed advanced algorithms and system designs that contributed to the successful deployment of the Iridium and Thuraya satellite communications networks, resulting in multiple issued patents. Earlier, as a Senior Staff Member at GTE Laboratories in Boston, Massachusetts, I conducted R&D in voiceband modem technologies and high-speed packet-switching systems. In parallel with industry leadership roles, I taught Electrical Engineering courses at Northeastern University and developed and delivered numerous advanced technical workshops for engineering and management teams across the telecommunications industry

Granted membership in the Motorola scientific and technical society for significant technical contributions, and received the Huawei certificate of recognition for the most outstanding individual and team leadership performance in AM KAD in 2011 (July 22)

Computer Tools/Standards Skills

Experience in Python programming and AI/ML frameworks, Strong expertise in extracting and analyzing network performance data from Business Objects and ENIQ platforms and developing effective root-cause analysis methodologies for network optimization. Proficient in a wide range of telecommunications planning, optimization, and performance management tools, including the Agilent RF analysis tool, TEMS Investigation, TEMS Cell Planner (TCP), NetAct, Planet, MapInfo, Ericsson ESPA, Astellia, SPW, Business Objects, and MATLAB. Demonstrated expertise in GSM, GPRS, UMTS, and 3G network planning, performance analysis, troubleshooting, and optimization using industry-standard methodologies and toolsets. Possess comprehensive knowledge of 3GPP specifications and standards for GSM and 3G technologies, complemented by hands-on experience with both Nokia and Ericsson network equipment and solutions. In addition, have a solid working knowledge of IEEE 802.16e wireless broadband technologies and WiMAX systems, enabling effective evaluation, design, and optimization of diverse wireless communication networks

Publications, Presentations and internal documents

"From LTE to LTE-Advanced Pro and 5G", with Marcin Dryjanski, Artechouse/USA, published September 2017

"UMTS Network Planning, Optimization, and Inter-operation with GSM", John Wiley & Sons, coming out late December 2007

"Smart proposed System design and KPI Documentation", Prepared as Internal document for Telcel/ Ericsson/Mexico, March 25/2008. Provides the specifications for an internal UMTS network performance analysis and optimization tool

"Mitigating interference between carriers", Wireless Asia, December 2005
"The Smart way of Planning UMTS in GSM Networks", Land Mobile Magazine, March 2006

"Interference and Coverage-Capacity Interactions in UMTS and the Implications on Network Planning", submitted to IEEE Wireless Comm Magazine for Pub, Oct, 2005 Oct, 2005

"Preparatory Steps for UMTS overlay in GSM networks", T-Mobile, internal document, August, 2005

"Overview of GSM Systems and Protocol Architecture", IEEE Communication Magazine, April 1993 (later became a Quarterly Reprint of IEEE publications on GSM)

“UMTS RF/radio planning guidelines”, T-Mobile, internal document, Nov 2005

“multi-operator interference management”, submitted for pub, Land Mobile, Oct 2005

“UMTS Antenna Integration with existing GSM sites”, T-Mobile, internal document, Oct 2005

www.lcc.com/media_center/publications.htm , pointer to several of my white papers on LCC website

“Design Considerations for Core Networks in GPRS and UMTS”, 3GIS Annual conference, Athen/Greece, July 2-3/2001.

“The Challenges of implementing and operating GPRS through the existing GSM network infrastructures”, presented at the IIR Telecoms & Technology Conference, London, Sept 2001

“A vision for Optical Core Networks in Future 3G Systems”, IP_Core Network Analyst, Oct 2001
“3G---Via Optical Core Networking”, Electronics World, January 2002

“Optimum soft Decision decoding with channel state information in the presence of fading”, IEEE Communication Magazine, Vol. 35, No.7, pp 110-111, July 1997 (with Antia,Y.)

"Frame Relaying and the Fast Packet Switching Concepts and Issues", IEEE Network Magazine, July 1991.
"The Fast Packet Ring Switch: A High Performance Efficient Architecture with Multi-cast Capability", IEEE Transactions on Communications, Vol 38, No 4, April 1990

"Channel equalization for the GSM system", "International Journal of Wireless Information Networks, vol. 3,No. 3, 1996.
"Channel equalization for the GSM system", "International Journal of Wireless Information Networks,vol. 3,No. 3, 1996.

List of Patents:

- · A " Block phase estimator for the coherent detection of non-differentially phase modulated data bursts on Rician fading channels”, IJWIN, vol 6, No 3, 1999.
- · "Methods and apparatus for demand-assigned reduced rate out-of-band signaling channel", US patent 5465253, issued on 11/7/1995.
- · "Method and apparatus for adaptive directed route randomization and distribution in a richly connected communication network", US patent 5430729, issued on 07/04/1995

- · "Symbol timing recovery and tracking method for burst-mode digital communications",Us patent 5870443 issued on 2/9/99.
- · “Method and apparatus for optimum soft-decision Viterbi decoding of convolutional-differential encoded QPSK data in coherent detection”, US patent 5754600, issued 5/7/98.
- · A " Block phase estimator for the coherent detection of non-differentially phase modulated data bursts on Rician fading channels, patent issued on 6/6/00.
- · "Burst format and associated signal processing to improve frequency/timing estimation for random access", joint patent issued on 4/16/02.
- · "Channel Estimation under Fading", Patent issued on 1/30/01.
- · "Signaling maintenance for discontinuous information communication", joint patent issued on 5/07/02.
- · "Optimum Soft Decision Decoding with Channel State Information in the Presence of Fading", (authored jointly), IEEE Communications Magazine, July 1997.
- · "Channel equalization for the GSM system", "International Journal of Wireless Information Networks,vol. 3,No. 3, 1996.